

Lecture 3: Logistic Regression

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3.1 Example: Sinking Putts

3.1.1 Problem Setup

Question: Predict the **probability** that a putt is sunk as a function of the distance to the hole.

Data: 266,846 first putts from PGA Tour ShotLink data, including distance to the hole and whether the putt was sunk or not. Each row in the dataset is a single putt and includes the following variables:

- i : index of the i^{th} putt in our dataset
- y_i : 1 if the i^{th} putt was sunk, 0 otherwise
- x_i : distance to the hole of the i^{th} putt, measured in feet

Visualization: The first thing we should do is plot the data, which we do in Figure 3.1. What do you notice?

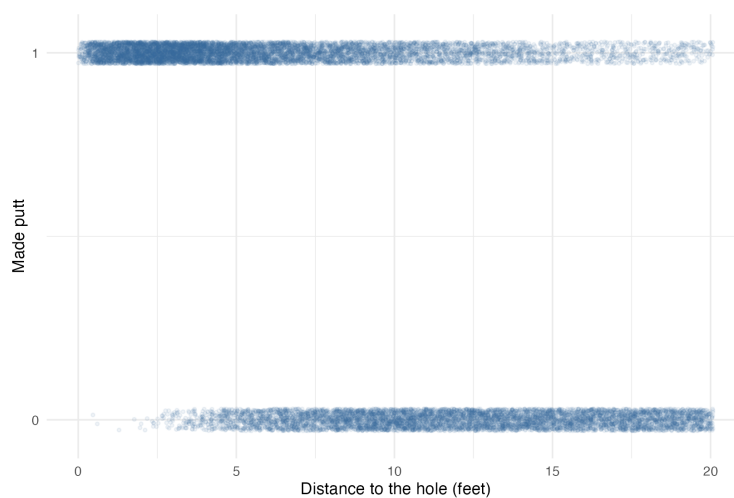


Figure 3.1: Scatter plot of the putt data

Like we would expect, the observed proportion of putts sunk decreases as the distance to the hole increases. Let's try to use what we've learned about linear regression to model this relationship.

3.1.2 Modeling

Model 1: Simple Linear Regression

We know from Lecture 1 that this model takes the form:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \mathbb{E}[\epsilon_i] = 0$$

We already know how to estimate the coefficients for this model. Recall that the least squares solution to this model is given by:

$$\begin{aligned} \hat{\beta}_0 &= \bar{Y} - \hat{\beta}_1 \bar{X} \\ \hat{\beta}_1 &= \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} \end{aligned}$$

We fit this model in R with the following code:

```
model_1 = lm(putt_made ~ distance, data = putt_data)  
summary(model_1)
```

When we plot the fitted line against the data in Figure 3.2, we see that the model does not fit the data well.

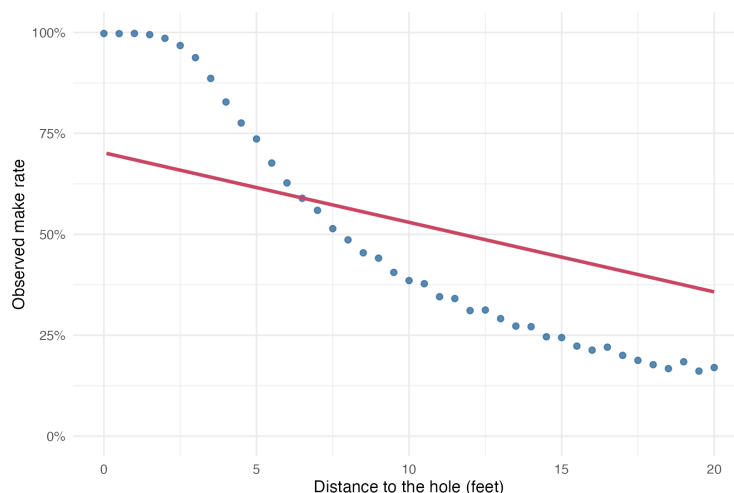


Figure 3.2: Putt data and fitted simple linear regression

The issue here is that the relationship is not close to linear in raw distance. The make rate drops very quickly for short putts, then flattens out. A more natural transformation is to work with $\log(x_i)$ instead of x_i .

Model 2: Linear Regression with a Log-Distance Transformation

This transformed model takes the form

$$y_i = \beta_0 + \beta_1 \log(x_i) + \epsilon_i, \quad \mathbb{E}[\epsilon_i] = 0$$

We estimate the coefficients in exactly the same way as before. The only thing that changes is the predictor we feed into the model. In matrix notation, the least-squares solution is still

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

We fit this model in R with the following code:

```
model_2 = lm(putt_made ~ log_distance, data = putt_data)
summary(model_2)
```

When we plot the fitted line against the data in Figure 3.3, we see that the transformed model fits the short-putt region much better.

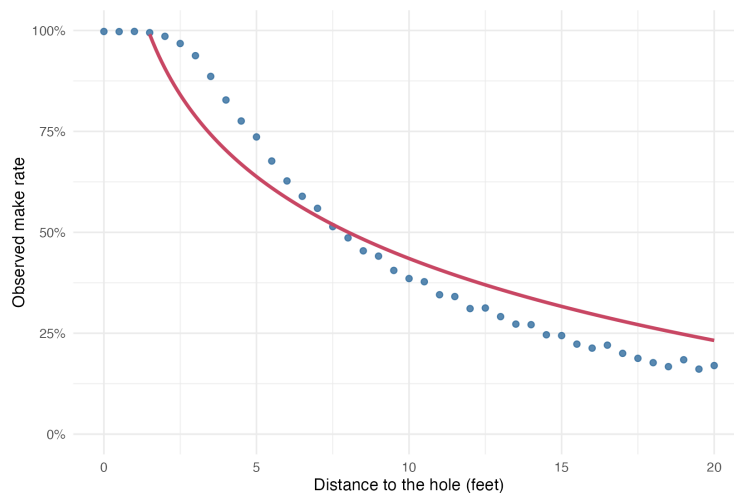


Figure 3.3: Putt data and fitted linear regression on log distance

However, when we zoom out in Figure 3.4, we still run into the same core problem: this is ordinary linear regression, so the fitted values are not constrained to stay in $[0, 1]$.

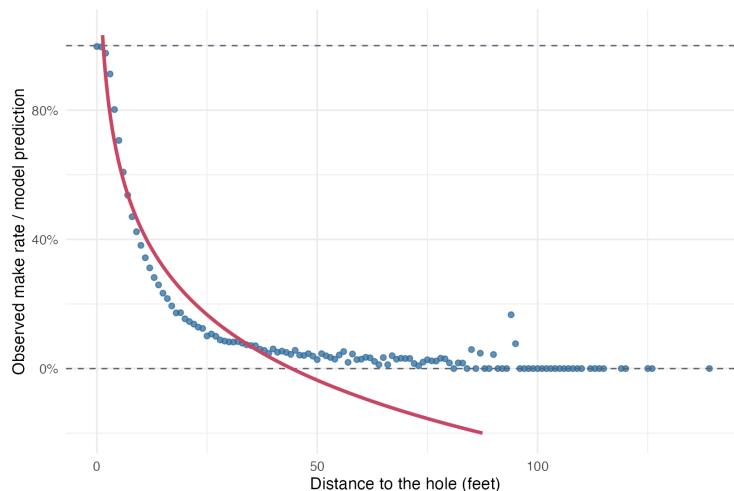


Figure 3.4: Putt data and fitted linear regression on log distance (zoomed out)

This presents two problems. First, we need our fitted probabilities to be between 0 and 1, but ordinary linear regression does **not** guarantee this. Second, binary outcomes do not naturally follow a constant-variance additive-error model: if $Y_i \sim \text{Bernoulli}(p_i)$, then $\text{Var}(Y_i | x_i) = p_i(1 - p_i)$ depends on the probability. How

can we constrain our model to ensure that our predictions are in $[0, 1]$?

3.2 Logistic Regression

3.2.1 The Logistic Function

We are looking for a "squishification" function that takes numbers in $\mathbb{R}$ and maps them to numbers in $(0, 1)$. One such function is the **logistic (or sigmoid) function**:

Definition 3.1 (Logistic Function). *The **logistic function** is defined as:*

$$\text{Logistic}(z) = \text{sigmoid}(z) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

We plot the logistic function in Figure 3.5.

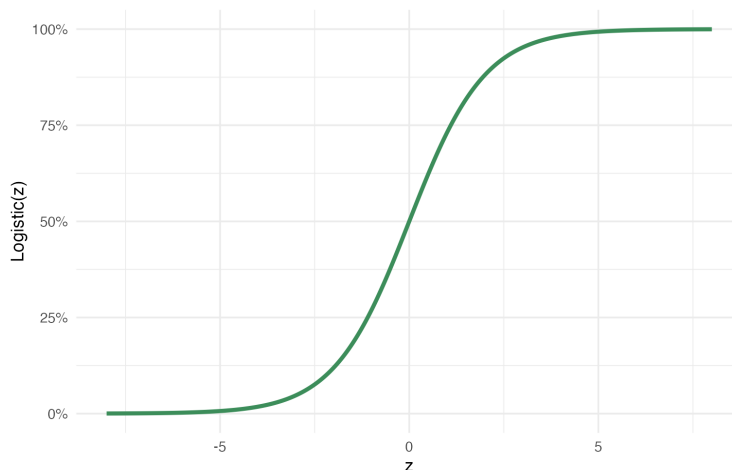


Figure 3.5: The Logistic Function

How can we implement this in our regression modeling?

3.2.2 The Model

Before, we had a linear predictor of the form

$$\eta_i = \beta_0 + \beta_1 z_i$$

where z_i is whichever transformed predictor we want to use. Here, a good choice is $z_i = \log(x_i)$. Now, we modify this model so that our fitted probabilities are between 0 and 1. We do this by making the conditional probability p_i the output of the logistic function:

$$p_i = \text{Logistic}(\eta_i) = \text{Logistic}(\beta_0 + \beta_1 z_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 z_i)}}$$

This is a **logistic regression model**, and allows us to model the **probability** that a putt is sunk directly. The outcome y_i is still binary; what the model outputs is the conditional probability that $y_i = 1$. Mathematically, we can write this model as:

$$p_i = \mathbb{P}(y_i = 1 \mid x_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 z_i)}}$$

$$\text{where } y_i \sim \text{Bernoulli}(p_i) = \begin{cases} 1 & \text{w.p. } p_i \\ 0 & \text{w.p. } 1 - p_i \end{cases}$$

3.2.3 Estimating the Coefficients

Our data is in terms of $y_i \in \{0, 1\}$ and $x_i \in \mathbb{R}$, not in terms of $p_i \in [0, 1]$. How can we then estimate the coefficients β in logistic regression?

In linear regression, we estimate β by minimizing the Residual Sum of Squares (RSS), or the squared error.

Definition 3.2 (Residual Sum of Squares). *The **Residual Sum of Squares** (RSS) is defined as:*

$$RSS(\beta) = \sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_p X_{ip}))^2$$

In logistic regression, we estimate β by minimizing the **log loss**, or the cross-entropy loss.

Definition 3.3 (Log Loss). *The **log loss** is defined as:*

$$L(\beta) = -\frac{1}{n} \sum_{i=1}^n (y_i \log p_i + (1 - y_i) \log(1 - p_i))$$

$$\text{where } p_i = \mathbb{P}(y_i = 1 \mid x_i, \beta) = \frac{1}{1 + e^{-(x_i^T \beta)}}$$

1. If $y_i = 1$, then observation i contributes $-\log p_i$ to $L(\beta)$
 - If $p_i \approx 1$, then $\log p_i$ is close to 0 and the loss contribution is low
 - If $p_i \approx 0$, then $\log p_i$ is very negative and the loss contribution is high
2. If $y_i = 0$, then observation i contributes $-\log(1 - p_i)$ to $L(\beta)$
 - If $p_i \approx 1$, then $\log(1 - p_i)$ is very negative and the loss contribution is high
 - If $p_i \approx 0$, then $\log(1 - p_i)$ is close to 0 and the loss contribution is low

The log loss is **convex** in β . Under regularity conditions, it has a unique finite minimizer.

$$\nabla_{\beta} L(\beta) = -\frac{1}{n} \sum_{i=1}^n (y_i \nabla_{\beta} \log p_i + (1 - y_i) \nabla_{\beta} \log(1 - p_i))$$

$$= 0$$

There is no closed-form least-squares-style solution. In practice, software finds the maximum-likelihood estimate numerically. In R, the `glm(..., family = "binomial")` function uses an iteratively reweighted least-squares algorithm.

3.3 Back to Sinking Putts

Recall the variables we have:

- i : index of the i^{th} putt in our dataset
- y_i : 1 if the i^{th} putt was sunk, 0 otherwise
- x_i : distance to the hole of the i^{th} putt

Since our transformed predictor is $\log(x_i)$, we also need $x_i > 0$, which holds for all putts in this dataset.

3.3.1 Fitting the Model

We use the `glm` function to fit the logistic regression model.

```
model_3 = glm(putt_made ~ log_distance ,  
              data = putt_data, family = "binomial")  
summary(model_3)
```

We plot the fitted line against the data in Figure 3.6. It seems to fit the data well, but we should check the model's performance from longer distances.

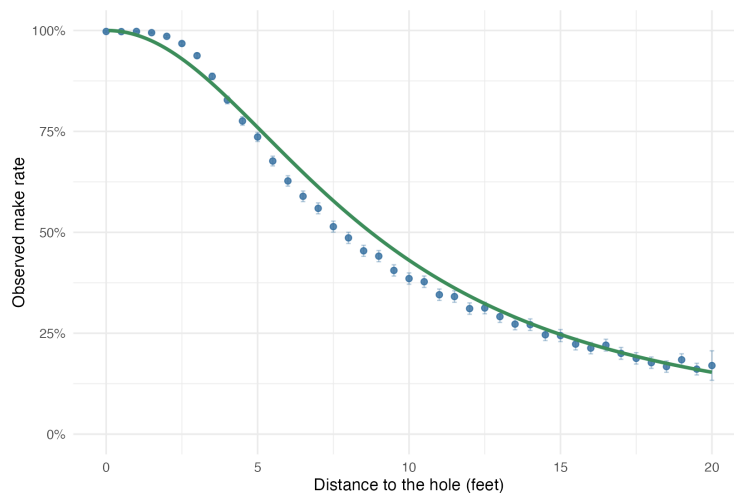


Figure 3.6: Putt data and fitted logistic regression

In Figure 3.7, we see that the fitted probabilities remain between 0 and 1 even beyond 20 feet. That is an important improvement over linear regression, although extrapolation is still model-based and should be checked against longer-distance data.

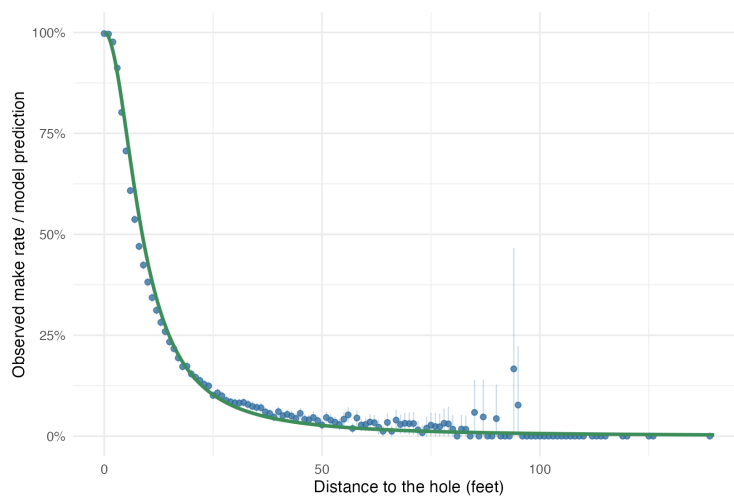


Figure 3.7: Putt data and fitted logistic regression (zoomed out)

3.3.2 Uncertainty in Logistic Regression

As in linear regression, our fitted coefficients are estimates, not ground truth. If we collected a new dataset of putts and refit the model, we would get slightly different values of $\hat{\beta}_0$ and $\hat{\beta}_1$.

In logistic regression, the coefficient estimates are approximately normal in large samples:

$$\hat{\beta} \approx N\left(\beta, (\mathbf{X}^T \widehat{\mathbf{W}} \mathbf{X})^{-1}\right),$$

where $\widehat{\mathbf{W}}$ is a diagonal matrix with entries

$$\widehat{W}_{ii} = \hat{p}_i(1 - \hat{p}_i).$$

Thus, the estimated variance-covariance matrix of $\hat{\beta}$ is

$$\widehat{\text{Var}}(\hat{\beta}) = (\mathbf{X}^T \widehat{\mathbf{W}} \mathbf{X})^{-1}.$$

The standard error of $\hat{\beta}_j$ is the square root of the j th diagonal entry:

$$SE(\hat{\beta}_j) = \sqrt{\left[\widehat{\text{Var}}(\hat{\beta})\right]_{jj}}.$$

An approximate 95% confidence interval for β_j is

$$\hat{\beta}_j \pm 1.96SE(\hat{\beta}_j).$$

In R, these coefficient estimates and uncertainty summaries are shown by:

```
summary(model)
confint(model)
```

Since logistic regression coefficients are on the log-odds scale, a confidence interval for β_j is a confidence interval for the change in log odds associated with a one-unit increase in predictor j , holding the other predictors fixed. If we exponentiate the coefficient, we get an odds ratio:

$$e^{\beta_j}.$$

So exponentiating a confidence interval for β_j gives a confidence interval for the corresponding odds ratio.

Now suppose we want uncertainty for the fitted probability at a new distance x_0 . Let

$$z_0 = \log(x_0), \quad \hat{\eta}_0 = \hat{\beta}_0 + \hat{\beta}_1 z_0,$$

so that

$$\hat{p}_0 = \frac{1}{1 + e^{-\hat{\eta}_0}}.$$

The standard error of the fitted log odds is

$$SE(\hat{\eta}_0) = \sqrt{\mathbf{x}_0^T \widehat{\text{Var}}(\hat{\boldsymbol{\beta}}) \mathbf{x}_0},$$

where $\mathbf{x}_0 = (1, z_0)^T$. An approximate 95% confidence interval for the log odds is

$$\hat{\eta}_0 \pm 1.96 SE(\hat{\eta}_0).$$

To get a confidence interval for the probability, we transform both endpoints back through the logistic function:

$$\left[\frac{1}{1 + e^{-(\hat{\eta}_0 - 1.96 SE(\hat{\eta}_0))}}, \frac{1}{1 + e^{-(\hat{\eta}_0 + 1.96 SE(\hat{\eta}_0))}} \right].$$

This interval describes uncertainty in the fitted mean probability. It is the logistic-regression version of a confidence interval for the mean response.

Finally, there is still randomness in the outcome of one new putt. Even if we knew the true probability p_0 , the outcome would still be random:

$$Y_0 | x_0 \sim \text{Bernoulli}(p_0).$$

Equivalently, we can write

$$Y_0 = p_0 + \epsilon_0,$$

where

$$E[\epsilon_0 | x_0] = 0, \quad \text{Var}(\epsilon_0 | x_0) = p_0(1 - p_0).$$

So unlike ordinary linear regression, logistic regression does not have one constant error variance σ^2 . The outcome variance depends on the probability itself: it is largest near $p_0 = 0.5$ and smallest near $p_0 = 0$ or $p_0 = 1$.

After fitting the model, we can compute residuals such as

$$r_i = y_i - \hat{p}_i.$$

These residuals help us check whether the model is systematically overpredicting or underpredicting in certain regions, such as very short or very long putts.

3.3.3 Improving the Model

We can do even better by letting the log odds be a quadratic function of log distance. Our new model is

$$\mathbb{P}(y_i = 1 \mid x_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \log(x_i) + \beta_2 [\log(x_i)]^2)}}$$

We fit this model in R with the following code:

```
model_4 = glm(putt_made ~ log_distance + I(log_distance^2),
              data = putt_data, family = "binomial")
summary(model_4)
```

We plot the fitted line against the data in Figure 3.8, and we can see that it tracks the observed make-rate curve more closely while still respecting the $[0, 1]$ constraint.

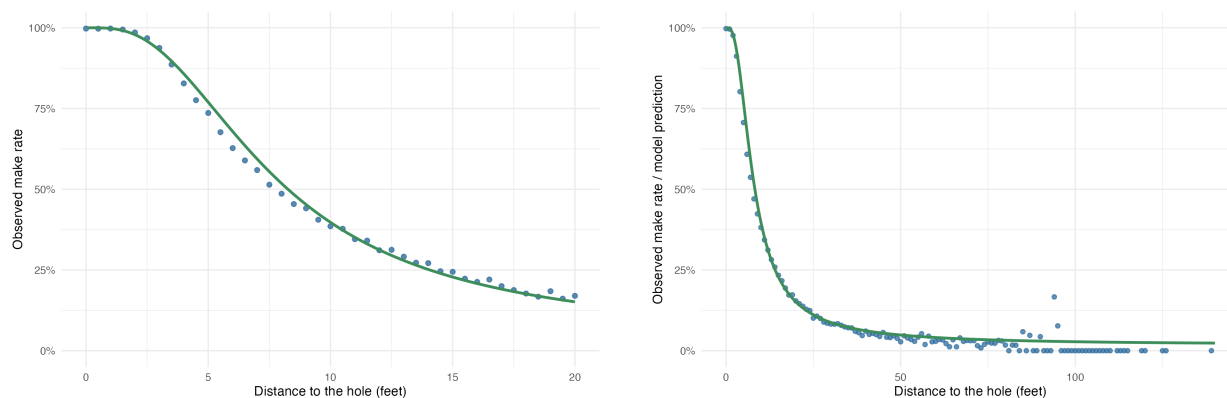


Figure 3.8: Putt data and fitted logistic regression with a quadratic in log distance

3.3.4 Takeaways

What can we take away from this example?

- Use linear regression to predict a value that can take any real number
- Use logistic regression to predict a **probability** in $(0, 1)$
- In logistic regression, we model the conditional probability $\hat{p}_i = \mathbb{P}(Y_i = 1 \mid x_i)$ rather than the binary outcome itself
- Coefficients in logistic regression describe how predictors affect the **log odds**, holding other predictors fixed
- When transformed or polynomial terms are present, coefficient interpretation requires care. For example, in the final model the effect of log distance on the log odds is $\beta_1 + 2\beta_2 \log(x_i)$, not just β_1 .